THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 11 & 12: **Keeping Time with the Skies, How Nature Works in Harmony** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The time between successive spring equinoxes is the: 1

(a) leap year **(b)** sidereal year **(c)** tropical year **(d)** lunar year

2. The first month of the Indian National Calendar is: 1

(a) Chaitra **(b)** Magha **(c)** Asadha **(d)** Vaisakha

3. Sun rises exactly in the east on: 1

(a) all days **(b)** new moon days **(c)** equinoxes **(d)** solstices

4. Cartosat satellites are used for: 1

(a) cooking **(b)** mapping and Earth observation

(c) tasting food **(d)** weather balloons only

5. The Surya Siddhanta is an: 1

(a) river **(b)** ancient Indian astronomical text

(c) modern calendar **(d)** satellite

6. In mutualism: 1

(a) both organisms benefit **(b)** one benefits and one is harmed

(c) none benefits **(d)** one is unaffected

7. Ecosystem services include: 1

(a) highways

(b) mines

(c) factories and cars

(d) clean air, water, food and climate regulation

8. Kunapa Jala in Vrikshayurveda is a: 1

(a) pesticide

(b) seed

(c) liquid fertiliser from animal and plant waste

(d) weedicide

9. Which is a major threat to ecosystems? 1

(a) planting trees **(b)** afforestation **(c)** deforestation **(d)** composting

10. The correct sequence for energy flow is: 1

(a) herbivore → producer → carnivore **(b)** producer → herbivore → carnivore

(c) decomposer → producer → sun **(d)** carnivore → herbivore → producer

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Name two festivals that follow solar sidereal calendars. 2

12. Why does the Moon rise later each day? 2

13. Why are mangroves in Sundarbans important? 2

14. Name two types of protected areas in India. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is a new moon? When can we see the crescent? 3

16. Describe how predators like tigers depend on prey. 3

17. Why were star positions at sunset, such as Makar near the winter solstice, recorded in ancient Indian texts like the Surya Siddhanta? 3

OR

How do satellites help in weather forecasting and disaster management? 3

18. Why should local communities participate in conserving ecosystems? 3

OR

State three ways in which students can help conserve ecosystems. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Discuss the importance of satellites for weather, communication and mapping in India. 5

OR

Compare the Gregorian calendar and the Indian National Calendar in a table of five points. 5

20. Compare terrestrial and aquatic ecosystems with examples, abiotic factors and organisms. 5

OR

Draw and explain a grassland food web containing grass, grasshopper, mouse, frog, snake and hawk, and say what happens if snakes are removed. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND